

Esteban Gonzalez

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EDUCATION

Harvey Mudd College

Bachelor of Science, Engineering

Cumulative GPA: 3.42/4.00 | Honors: Dean's List

Claremont, CA

Graduation Date: May 2028

SKILLS

Languages/Software: C++, Python, MATLAB, LabVIEW, SolidWorks, GNU Radio, Radiant, Microsoft Office (Excel, Word, PowerPoint)

Lab Equipment: Network analyzers, oscilloscopes, software-defined radios (SDR)

EXPERIENCE

Aerospace Corporation / FAA (Harvey Mudd Clinic)

Claremont, CA

Software-Defined Radio Team Lead

Aug 2026 – Dec 2026

- Led RF characterization of a Controlled Reception Pattern Antenna (CRPA) using a USRP Software-Defined Radio (SDR), designing custom test procedures for Signal-to-Noise Ratio (SNR), relative phase and amplitude, null-forming, and beam steering
- Engineered and executed RF measurement protocols in an anechoic chamber utilizing a Vector Network Analyzer (VNA) and SDR hardware to evaluate antenna anti-jamming capabilities
- Automated spatial data collection using motorized positioners in an anechoic chamber to generate high-resolution 3D radiation pattern plots for antenna performance validation

LILAQ Lab Research – Harvey Mudd College

Claremont, CA

Undergraduate Researcher

Jan 2026 – Present

- Engineered low-cost indoor air quality filtration prototypes based on Corsi-Rosenthal box mechanics, utilizing stacked PC fan arrays to maximize airflow and minimize acoustic noise
- Benchmarked prototypes against commercial air purifiers (\$168–\$390 range); achieved 176 CFM Clean Air Delivery Rate (CADR) at 39 dB for \$163, outperforming market standards (e.g., Coway Airmega 150 at 161 CFM, 40 dB, \$190) across cost, noise level, and clean air output

Engineering Department – Harvey Mudd College

Claremont, CA

Tutor and Grader, Digital Electronics and Computer Architecture

Aug 2026 – Dec 2026

- Provide 1-on-1 instruction and mentorship to 20+ students in a 100-student digital design course, breaking down concepts in C programming, assembly language, embedded systems, and microarchitecture

PROJECT EXPERIENCE

Autonomous Surface & Dive Robot

Claremont, CA

Team Lead – Harvey Mudd College Engineering Systems

Jan 2026 – May 2026

- Programmed embedded Teensy firmware to control sonar burst timing and sample sensor data at 10 Hz, meeting Nyquist requirements for acoustic time-of-flight range measurements
- Processed field data in MATLAB to reconstruct harbor floor depth profiles, validating results against Dana Point Harbor Authority survey charts, and iterated the frame design in SolidWorks to resolve diving instability found in tank testing
- Designed and calibrated analog sensor circuits (thermistor, pressure, sonar transceiver) for an AUV, achieving depth accuracy within 0.1 m and temperature accuracy within 0.1°C of reference instruments

RISC-V Multicycle Processor

Claremont, CA

Individual Project

Oct 2025 – Dec 2025

- Architected and implemented a 32-bit RISC-V microprocessor using SystemVerilog, supporting the core RV32I base integer instruction set specification
- Executed functional verification in Siemens QuestaSim, developing custom testbenches and analyzing signal waveforms to validate instruction execution, datapath control, and timing accuracy